In this case, the experiments were done using the learning rates of 0.001, 0.01 and 0.1. All three learning rates converged similarly and performed almost equally well. The learning rate impacted mainly the size of weight changes rather than affecting the accuracy of classification.

4. Explain why a Single Layer Perceptron cannot solve the XOR problem.

   Single Layer Perceptron can classify only linearly separable data. As the XOR problem is not linearly separable, a single decision boundary cannot distinguish its classes. Hence, Multilayer Perceptron is needed for solving the XOR problem.

5. Study the effect of feature normalization on model convergence.

   Feature scaling will ensure that all input features have a standard scale value, thereby avoiding the domination of any feature with higher value. In the current experiment, MinMax scaling will increase training stability and allow for faster convergence of the perceptron.

10. Conclusion

In the experiment, the Single Layer Perceptron was successfully developed for the purpose of classifying the Banknote Authentication dataset into two classes. For this, the dataset was normalized through MinMax technique while the perceptron learning algorithm and step activation function were used to train the classifier. As a result, the classifier performed extremely well on linearly separable data.

11. References

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